Advanced Fiber Materials (2026) 8:577–593
https://doi.org/10.1007/s42765-025-00633-7

RESEARCH ARTICLE

Nanotextured Tribonegative PTFE-SiO_2 Submicron Fibers for Self-Cleanable and Highly Flexible Kinetic Energy Harvester

Freddy Baltazar Iniguez[1,2] · Daekyu Choi[1,2] · Peijin Jiang[1,2] · Dogun Park[1,2] · Kiyong Kim[1,2] · Min Wook Lee[3] · Hongseok Jo[1,4] · Seongpil An[1,2,5,6]

Received: 14 June 2025 / Accepted: 9 October 2025 / Published online: 3 November 2025
© Donghua University, Shanghai, China 2025

Abstract

The triboelectric nanogenerator (TENG) has emerged as a promising renewable energy technology for harvesting kinetic energy from natural sources, such as human motion and rainfall. In this study, we fabricate a mechanically robust, porous, and superhydrophobic PTFE-SiO_2 nanofiber (NF)-based TENG using a facile electrohydrodynamic spinning technique, *i.e.*, electrospinning method. The incorporation of SiO_2 nanoparticles (NPs) into the PTFE NF matrix significantly enhances the output performance, energy-harvesting efficiency, and superhydrophobic characteristics of the PTFE-SiO_2 NF-based TENG. The optimized PTFE-SiO_2 NF-based TENG achieves a maximum energy-harvesting efficiency of 102 $mW{\cdot}N^{-1}{\cdot}m^{-2}$, outperforming previously reported NF-based TENGs. Moreover, it successfully harvests kinetic energy from multiple natural stimuli, generating 429 μW from human interaction and 1.65 μW from water droplets. These results demonstrate the potential of PTFE-SiO_2 NF-based TENG for integration into self-powered wearable electronics and environmental energy-harvesting systems for their autonomous operation without reliance on external power sources.

Keywords Triboelectric nanogenerator · Kinetic energy-harvesting · Superhydrophobicity · Electrospinning · PTFE-SiO_2 nanofiber mat

1 Introduction

The rapid advancement of wearable electronics, autonomous sensors, and human–machine interfaces has intensified the demand for lightweight, flexible, and self-powered energy sources [1–4]. Among the various emerging energy-harvesting technologies, triboelectric nanogenerators (TENGs) have garnered significant attention due to their high energy conversion efficiency, structural simplicity, and broad adaptability in converting mechanical stimuli into electricity [5–8]. TENGs exploit the coupling of contact electrification and electrostatic induction to harvest kinetic energy from various ambient sources, including human motion, air flows, water waves, and others [9–11]. To enhance their electrical output and facilitate practical applicability, extensive research has focused on optimizing the selection of triboelectric

Freddy Baltazar Iniguez and Daekyu Choi have contributed equally to the work.

✉ Min Wook Lee
mwlee0713@kist.re.kr

✉ Hongseok Jo
johs@jbnu.ac.kr

✉ Seongpil An
esan@skku.edu

[1] SKKU Advanced Institute of Nanotechnology (SAINT), Sungkyunkwan University (SKKU), Suwon 16419, Republic of Korea

[2] Department of Nano Science and Technology, Sungkyunkwan University (SKKU), Suwon 16419, Republic of Korea

[3] Institute of Advanced Composite Materials, Korea Institute of Science and Technology, Chudong-ro, Bongdong-eup, Wanju-gun, Jeonbuk 55324, Republic of Korea

[4] Department of Mechanical Engineering, Jeonbuk National University, 567 Baekjedae-ro, Deokjin-gu, Jeonju-si, Jeonbuk 54896, Republic of Korea

[5] Department of Nano Engineering, Sungkyunkwan University (SKKU), Suwon 16419, Republic of Korea

[6] Department of Semiconductor Convergence Engineering, Sungkyunkwan University, Suwon 16419, Republic of Korea

materials, engineering advanced device architectures, and employing surface modification techniques to enhance the performance of TENGs. [12–14].

Polytetrafluoroethylene (PTFE), a fluorinated polymer with strong electron affinity, ranks among the most tribonegative materials in the triboelectric series (Fig. S1). Its inherent chemical inertness, thermal stability, low surface energy, and excellent dielectric properties make PTFE an attractive candidate for achieving robust and high-performance TENGs [15, 16]. However, conventional fabrication methods associated with nano-textured two-dimensional (2D) PTFE structures, including mechanical stretching and plasma etching, are often limited by their complexity, low scalability, and the need for expensive equipment [17, 18]. Furthermore, the inherently smooth surface of 2D PTFE can restrict the effective surface area, thereby impeding optimal charge generation.

To overcome these limitations, one of the electrohydrodynamic techniques, *i.e.*, electrospinning, has emerged as an effective and versatile strategy for fabricating sub-micron fiber-based 2D PTFE structures [19, 20]. Through emulsion or suspension electrospinning followed by thermal sintering, PTFE nanofibers (NF) with hierarchical micro- and nanostructures can be produced. These nano-textured PTFE NF mats exhibit significantly enhanced surface area, porosity, and mechanical flexibility compared to conventional flat PTFE films, resulting in a dramatic improvement in triboelectric performance. Lin et al. developed a self-powered sensor device based on PTFE NF that effectively converts mechanical energy into electrical energy, achieving a high output power of 56.25 μW [21]. This device demonstrated potential for applications in body motion detection and health monitoring. In addition, Zhao et al. reported a PTFE NF-TENG utilizing negatively charge-injected electrospun PTFE membranes, which achieved a peak-to-peak open-circuit voltage of approximately 900 V and a peak power density of 9 $W{\cdot}m^{-2}$ [22].

The introduction of composite strategies has further expanded the functional versatility of electrospun PTFE NF. Incorporating nanoscale additives such as silica (SiO_2), carbon nanotubes (CNTs), and other inorganic nanofillers enables the development of multifunctional triboelectric layers with tailored surface roughness, wettability, conductivity, and mechanical resilience. In particular, the addition of SiO_2 nanoparticles (NPs) in PTFE NF has been shown to impart superhydrophobic and self-cleaning properties to the resulting PTFE NF, effectively addressing water-induced performance degradation and fouling issues in humid environments. For example, Zou et al. successfully fabricated electrospun PTFE-SiO_2 NF membranes with excellent dynamic hydrophobicity, capable of resisting water droplet penetration at velocities up to 3.376 $m{\cdot}s^{-1}$ [23].

The PTFE-SiO_2 NF TENG achieved a maximum power density of 2.04 $W{\cdot}m^{-2}$, significantly surpassing previously reported nanofiller-based TENGs, including CNT/molybdenum disulfide (MoS_2)-polyvinylidene fluoride (PVDF) (134 $mW{\cdot}m^{-2}$), MWCNTs/polyacrylonitrile (PAN) (48 $mW{\cdot}m^{-2}$), polyethylene terephthalate (PET)-titanium dioxide (TiO_2)-nylon 6,6 (23.44 $mW{\cdot}m^{-2}$), and TiO_2/MXene–polystyrene (PS) (25.2 $\mu W{\cdot}cm^{-2}$) [24–27]. This superior performance highlights the benefits of SiO_2 over traditional fillers, such as CNTs and TiO_2, which are limited by leakage current or UV-dependent activation [28–30].

Our work presents a self-cleanable, wearable, and performance-improved TENG utilizing hydrophobic and nanoporous PTFE-SiO_2 NF for demonstrating potential as a water droplet energy-harvesting device. By integrating nanoscale roughness, porous network structures, and enhanced hydrophobicity, our design addresses key challenges in water-contact TENG applications, including performance instability, mechanical degradation, and poor environmental adaptability.

The PTFE-SiO_2 NF mat was fabricated using electrospinning followed by sintering, achieving an optimized balance between effective contact surface area, mechanical durability, and surface charge density. This study contributes to the development of a novel design for next-generation TENGs and presents a scalable and environmentally friendly approach for fabricating functional PTFE-based nanomaterials. The multifunctionality imparted by the PTFE-SiO_2 NF composite, including self-cleanability, flexibility, and water resistance, enables the proposed device to be readily integrated into smart textiles, outdoor sensing platforms, and self-powered environmental monitoring systems.

2 Experimental Section

2.1 Materials

Poly(ethylene oxide) PEO (M_v = 600 kDa), a polytetrafluoroethylene (PTFE) aqueous dispersion (60 wt% suspension in H_2O), LUDOX TM-50 colloidal silica (SiO_2, 50 wt% suspension in H_2O), nylon 6/6 (Polyamide 66), and formic acid (99%) were purchased from Sigma-Aldrich Ltd., Republic of Korea. All the chemicals were of analytical grade and used without additional purification. Deionized (DI) water was prepared in-house and was used for all the experiments.

2.2 Preparation of Electrospinning Solutions

To prepare the PTFE-SiO_2 solution, 5 wt% of PEO aqueous solution was initially prepared by dissolving PEO powder in DI water while stirring overnight, as reported previously [23]. Then, aqueous PTFE dispersion was added

to the PEO solution under continuous stirring, resulting in a concentration ratio of 95:5 (PTFE: PEO wt%). SiO_2 aqueous dispersion with various concentrations (4, 7, and 10 wt% vs. PTFE) was also introduced into the PTFE-PEO solution and stirred continuously at room temperature (cf. Fig. 1a). In addition, 8 wt% of nylon 6/6 solution was prepared by dissolving nylon 6/6 pellets in formic acid (99%) and stirring at room temperature for 24 h.

2.3 Fabrication of Nylon 6/6 and PTFE-SiO_2 Nanofiber Mats

For the fabrication of nylon 6/6 nanofiber (NF) mats, nylon 6/6 solution was electrospun using a syringe pump (Legato 210, KD Scientific, USA) equipped with a 10-mL syringe and 20-gauge needle. The solution was electrospun onto a copper foil with an applied DC voltage of 17 kV (FC20P6, Glassman High Voltage, USA). The flow rate and the distance between the needle and collector were fixed at

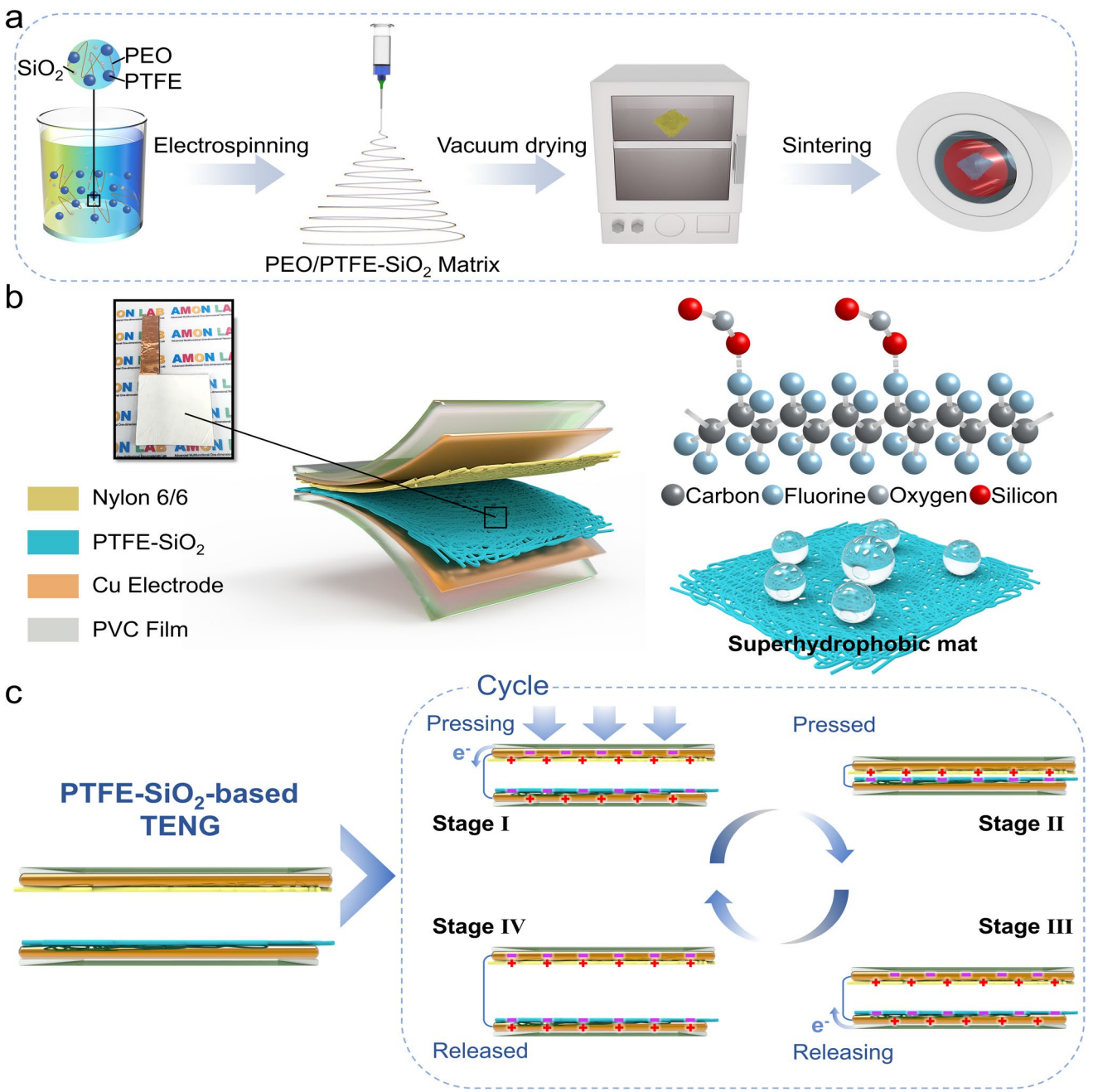


Fig. 1 Schematics of **a** the fabrication process, **b** the structure of the PTFE-SiO_2-based TENG, and **c** the working mechanism of the TENG

200 μL·h^{-1} and 10 cm, respectively. Electrospinning was performed at room temperature and a relative humidity of 30%. Pristine PTFE and PTFE-SiO_2 NF mats containing 4, 7, and 10 wt% of SiO_2 nanoparticles (NPs) were also fabricated via electrospinning using an 18-gauge needle onto a stainless-steel mesh (400 mesh) under an applied DC voltage of 7–8 kV and a flow rate of 500 μL·h^{-1}. A needle-to-collector distance was 10 cm. The electrospun NF mats were subsequently dried under vacuum for at least 8 h to remove residual solvents. Finally, the pristine PTFE and PTFE-SiO_2 NF mats were sintered at 380 °C for 10 min in ambient air (cf. Fig. 1a).

2.4 Fabrication and Energy-Harvesting Performance Testing of Triboelectric Nanogenerator

Figure 1a illustrates the fabrication process for triboelectric nanogenerator (TENG). Rectangular PTFE-SiO_2 NF and nylon 6/6 NF mats, having tribonegative and tribopositive properties, respectively, were adhered onto a Cu-electrode mounted on a polyvinyl chloride (PVC) film (Fig. 1b). A nanogenerator testing machine equipped with an oscilloscope (TBS2000B, Tektronix, USA), multimeter (DMM7510, KEITHLEY, USA), force gauge indicator (CL-600, CAS, Republic of Korea), and load cell (CBCA20L, Curiosity Technology, Republic of Korea) were employed to evaluate the energy-harvesting performance of TENGs. The size of each TENG was 40 mm × 40 mm. Note that the size of the PTFE-SiO_2 NF TENG for practical applications, including contact between human skin and water droplets, was 20 mm × 40 mm. The output voltage and current signals were acquired using a multimeter with an internal resistance of 10 MΩ and 100 Ω under the open- and short-circuit conditions, respectively.

2.5 Characterizations

The morphology of NF mats was characterized using a field-emission scanning electron microscope (FE-SEM, S-5200, HITACHI, Japan) with an accelerating voltage of 15 kV. The average diameter of NF was determined by measuring the diameter of 100 NFs using ImageJ software. Energy dispersive X-ray analysis (EDS) (JSM-IT800, JEOL, Japan) was used to analyze the elemental composition of the mat. X-ray diffraction (XRD, D8 ADVANCE, Bruker, USA) and Fourier-transform infrared (FTIR) spectroscopy (Horiba LabRam Aramis IR2, Japan) were utilized to analyze the crystalline structures and chemical bondings of the PTFE and PTFE-SiO_2 NF. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) (SDT-650, TA Instruments, USA) were conducted to identify the presence of PTFE and SiO_2 in the NF mats under nitrogen gas atmosphere with a temperature-rising rate of 10 °C/min from 25 to 700 °C.

Kelvin probe force microscopy (KPFM) using an atomic force microscope (AFM, NX10, Park Systems, Republic of Korea) with PPP-NCSTAu prove (tip dimension = 150 μm × 27 μm × 2.8 μm, force constant = 7.4 N/m, and resonance frequency = 160 Hz) was carried out to examine the charge potential and work function of the NF mats under a set scan speed of 0.5 Hz and a tip-to-specimen distance of 13.1 μm. The roughness of the NF mats was measured by 3D laser confocal microscope (3D LSM) using an optical microscope (OLS5100-SAF, LEXT, Japan). The water contact angle of NF mats was measured using a contact angle meter (SmartDrop, Femtobiomed, Republic of Korea). Tensile testing of NF mats (30 mm × 10 mm) was conducted using a universal testing machine (UTM, 34SC-05, Instron, USA) equipped with a static load cell of 100 N at a strain rate of 4 mm/min. Young's modulus (E) and yield stress (Y) were determined from the acquired data using the elastic–plastic Green's equation (Eq. 1) fitted to the stress–strain curves described in previously reported studies [31–33].

$$\sigma_{xx} = Y \tanh\left(\frac{E}{Y}\varepsilon_{xx}\right) \quad (1)$$

where σ_{xx} and ε_{xx} are the tensile stress and the corresponding tensile strain, respectively. The average values of E and Y were determined from three measurements.

3 Results and Discussion

3.1 Fabrication of PTFE-SiO_2 Nanofiber-Based Triboelectric Nanogenerator

The electrospun polytetrafluoroethylene (PTFE)-silica (SiO_2) NF mats were fabricated by varying the SiO_2 nanoparticle (NPs) concentrations from 0 to 10 wt% to investigate the effect of SiO_2 NPs on the morphology of the NF and the energy-harvesting performance of the triboelectric nanogenerator (TENG). Figure 1a and c illustrate the fabrication process and working mechanism of the TENG, respectively. The PTFE-SiO_2 NF mat and nylon 6/6 NF mat were employed as tribonegative and tribopositive materials, respectively (Fig. S1). The prepared NF mats were cut into 40 mm × 40 mm rectangles and attached to Cu foils that served as electrodes. These specimens were then adhered to a polyvinyl chloride (PVC) film and interconnected to form a closed-loop-shaped TENG (Fig. 1b) [34–38].

The PTFE-SiO_2 NF TENG harvests electrical energy through repeated contact and separation between the tribonegative PTFE-SiO_2 NF mat and the tribopositive Nylon 6/6 NF mat (Fig. 1c). The working mechanism of a TENG

is based on the principle of coupling triboelectrification and electrostatic induction [37–39]. The triboelectric effect originates from electron transfer, driven by the interaction between the overlapping electron clouds at the interface between two materials [35]. When two dissimilar materials come into contact (Stage I in Fig. 1c), electron transfer occurs because of differences in electron affinity and interfacial properties, such as chemical composition and surface roughness [40–42]. During compression (Stage II in Fig. 1c), opposite charges accumulate on the surfaces until electrostatic equilibrium is established. Upon separation (Stage III in Fig. 1c), an electrical potential difference forms, driving electron flow between the electrodes and generating an induced current when connected to an external circuit. By repeating the contact-separation process, the TENG can continuously generate an electrical output.

On the other hand, this study demonstrated the single-electrode mode operation of the PTFE-SiO_2 NF TENG, using the ground as the reference electrode, which facilitates energy harvesting in various scenarios involving human contact and water droplets. This mode demonstrated significant versatility, allowing energy harvesting from dynamic objects without requiring a direct electrical connection to a conductor. This capability made the PTFE-SiO_2 NF TENG suitable for diverse energy-harvesting applications, such as hand typing, human motion, raindrops, and transportation movement [42–46].

3.2 Mechanical and Physicochemical Characterizations of PTFE-SiO_2 Nanofibers

Figure 2 shows the field-emission scanning electron microscopy (FE-SEM) images of the electrospun nylon 6/6 and the PTFE-SiO_2 NF mats with varying SiO_2 contents. The nylon 6/6 NF with an average diameter (D_{avg}) of (104.8 ± 14.8) nm exhibited highly smooth surfaces (Fig. 2a and Table 1).

Significant morphological changes were observed in the PTFE-SiO_2 NF after the sintering process, with these changes dependent on the SiO_2 content. Before sintering, the PEO/PTFE-SiO_2 NF, composed of PTFE and SiO_2 NPs embedded within the PEO matrix, had D_{avg} values of 1164.7, 1144.1, 912.3, and 1071.5 nm for 0, 4, 7, and 10 wt% SiO_2, respectively (Fig. S2a and b). After sintering at 380 °C, the PTFE-SiO_2 NF developed rough and porous structures, accompanied by a decrease in D_{avg} compared to the pre-sintered NFs (Fig. 2b – e) [47]. This change was attributed to the decomposition of PEO at the sintering temperature of 380 °C. Furthermore, an increase in SiO_2 content led to decreased interconnectivity between the NFs due to the reduced concentration of PTFE, which limited sintering-induced fusion. The surface elemental composition of the PTFE–SiO_2 (7 wt%) NF mat was analyzed using energy-dispersive spectroscopy (EDS) (Fig. S2c). The detection of Si and O elements confirms the uniform distribution of SiO_2 NPs within the PTFE NF matrix.

Figure 3a-c present the stress–strain curves of nylon 6/6, pristine PTFE, and PTFE-SiO_2 (7 wt%) NF mats, while Fig. S3a and b show the stress–strain curves for PTFE-SiO_2 NF mats at 4 and 10 wt%. The nylon 6/6 NFs mat exhibited a maximum stress (σ_{max}) of (5.17 ± 0.27) MPa and a maximum strain (ε_{max}) of (22.2 ± 6.55)% (Fig. 3a). In contrast, the pristine PTFE NFs mat showed a lower σ_{max} of (3.72 ± 1.27) MPa but a much higher ε_{max} of (250 ± 10.7)% (Fig. 3b).

The incorporation of SiO_2 NPs (7 wt%) further reduced both σ_{max} and ε_{max} to (1.86 ± 0.35) MPa and (32.1 ± 5.47)%, respectively (Fig. 3c). At lower loading (4 wt%), σ_{max} and ε_{max} decreased sharply to (0.39 ± 0.09) MPa and (19.8 ± 2.35)%, respectively (Fig. S3a). At higher loading (10 wt%), σ_{max} partially recovered to (1.04 ± 0.51) MPa, whereas ε_{max} further decreased to (14.8 ± 1.50)% (Fig. S3b). This behavior can be attributed to SiO_2-induced variations in fiber diameter and inter-fiber fusion. At 4 wt%, the relatively larger fiber diameter reduces the specific surface area and weakens inter-fiber fusion (Fig. 2c), resulting in poor mechanical properties. At 7 wt%, the fibers reach their minimum diameter, which facilitates enhanced fusion and more effective stress transfer (Fig. 2d), thereby yielding the optimal balance between strength and elongation. However, further increasing SiO_2 loading to 10 wt% enlarges the fiber diameter again (Fig. 2e), which slightly reinforces the stress response via improved fusion but reduces strain capacity due to increased porosity and restricted deformability.

This reduction in mechanical properties was attributed to the formation of inherent defects, such as microcracks, resulting from the incorporation of brittle SiO_2 NPs into the PTFE matrix. Note that these defects act as crack initiation sites, which can cause further crack propagation during stretching [48]. However, the incorporation of SiO_2 NPs resulted in an increase in Young's modulus (E) while decreasing the yield stress (Y) (Table 1) compared to the pristine PTFE NF mat. This indicates enhanced stiffness and resistance to deformation. The reduction in yield stress implies that the presence of SiO_2 NPs restricted the relative displacement of the molecular chains within the PTFE matrix, thereby diminishing its elastic properties [49]. Consequently, the PTFE-SiO_2 NFs exhibited increased stiffness, albeit with reduced overall flexibility of the NF mats [50].

Physicochemical characterization techniques, including X-ray diffraction (XRD), differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), and Fourier-transform infrared (FTIR) spectroscopy, were used to analyze the chemical properties of the electrospun mats. Figure 3d shows the XRD patterns of the pristine PTFE and PTFE-SiO_2 NF mats. A prominent diffraction peak appears at $2\theta = 18.1°$, corresponding to the (100) crystal plane of

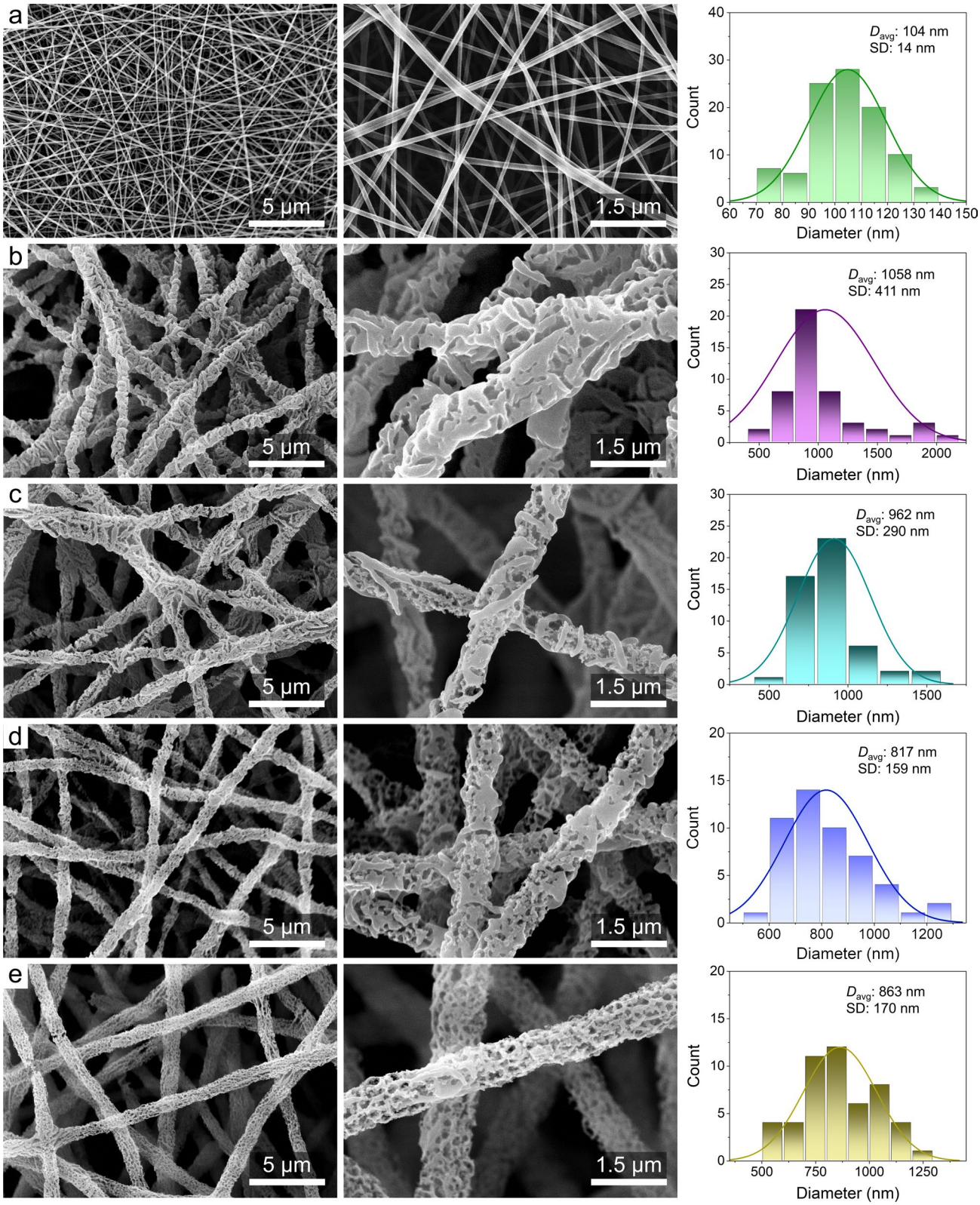


Fig. 2 SEM images of the NF mats and corresponding fiber size distribution: **a** pristine Nylon 6/6 and the sintered PTFE-SiO_2 NF mats with **b** 0, **c** 4, **d** 7, and **e** 10 wt%

Table 1 Mechanical properties of different electrospun NF mats

	Nylon 6/6	Pristine PTFE	PTFE-SiO_2 (4 wt%)	PTFE-SiO_2 (7 wt%)	PTFE-SiO_2 (10 wt%)
D_{avg} (nm)	104 ± 14	1058 ± 411	962 ± 290	817 ± 159	863 ± 170
E (MPa)	59.7 ± 20.7	5.35 ± 3.05	4.54 ± 0.19	59.8 ± 30.6	18.3 ± 0.51
Y (MPa)	4.25 ± 0.40	3.34 ± 1.14	0.24 ± 0.10	1.66 ± 0.49	0.81 ± 0.02
σ_{max} (MPa)	5.17 ± 0.27	3.72 ± 1.27	0.38 ± 0.09	1.86 ± 0.35	1.04 ± 0.51
ε_{max} (%)	22.2 ± 6.55	250 ± 107	19.8 ± 2.35	32.1 ± 5.47	14.8 ± 1.50

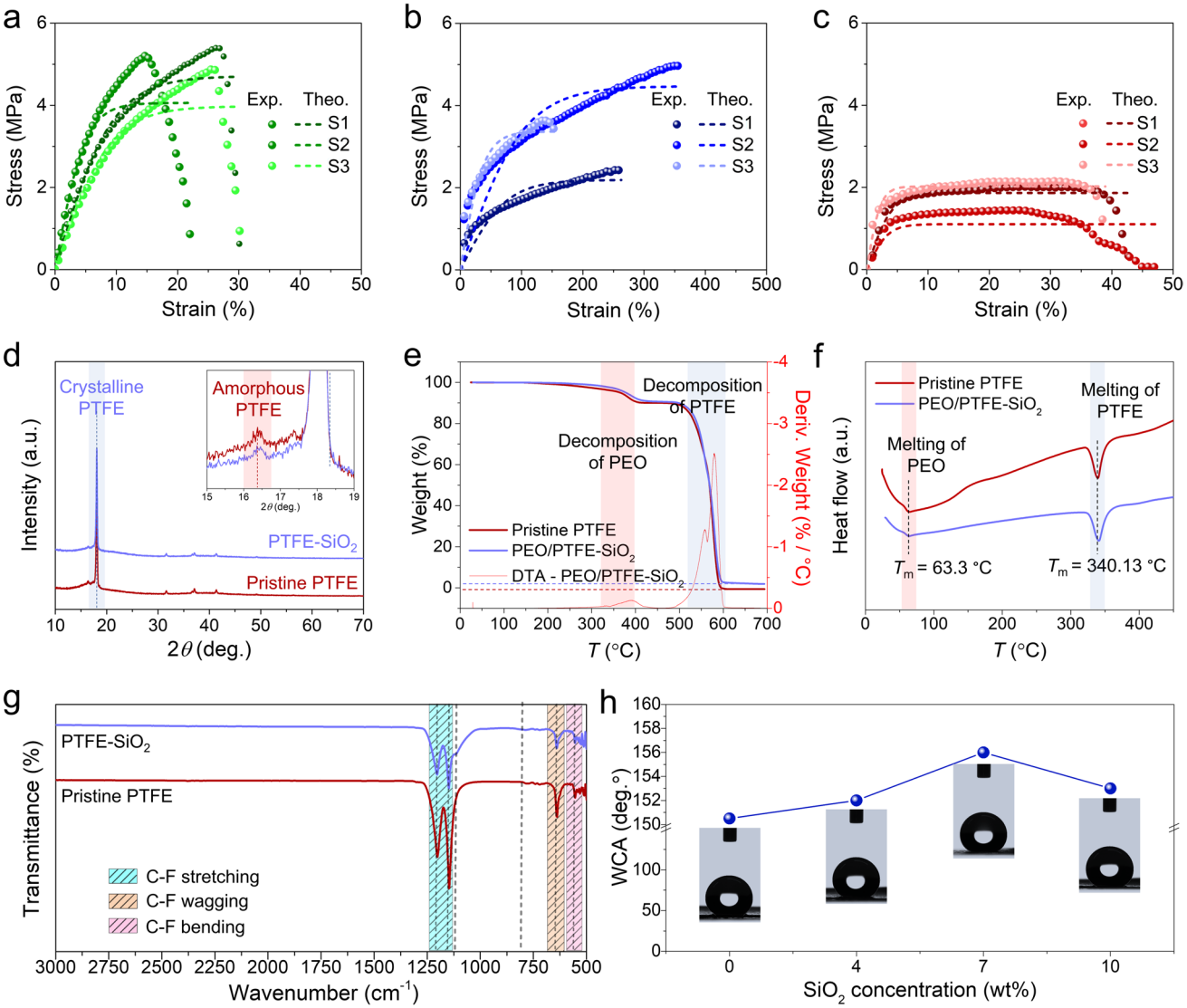


Fig. 3 Stress–strain curves of **a** nylon 6/6 NF mat, **b** Pristine PTFE NF mat, and **c** PTFE-SiO_2 (7 wt%) NF mat. **d** XRD patterns of pristine PTFE and PTFE-SiO_2 (7 wt%) NF mats. **e** TGA and **f** DSC curves of the pristine PTFE and PEO/PTFE-SiO_2 (7 wt%) NF mats. Note that DTA curve of the PEO/PTFE-SiO_2 (7 wt%) NF mat is also included in **(e)**. **g** FTIR spectra of the sintered pristine PTFE and PTFE-SiO_2 (7 wt%) NF mats. **h** WCAs of PTFE-SiO_2 NF mats with an increase in the SiO_2 content from 0 to 10 wt%

PTFE, along with smaller peaks at 31.62°, 37.1°, and 41.4°, associated with the (110), (107), and (108) crystal planes of PTFE, respectively [51–53]. A peak at $2\theta = 16.37°$ indicates the amorphous phase of PTFE [53]. Figure S4a shows a weak peak at $2\theta = 22°$, attributed to the amorphous SiO_2 (Fig. S4b), indicating a low SiO_2 content relative to PTFE.

Figure 3e and f show the TGA/DTA and DSC profiles of the pristine PTFE and PEO/PTFE-SiO_2 NF mats. In Fig. 3e, the TGA thermograms revealed a two-step mass loss process: the first at 380 °C due to the decomposition of PEO (see Fig. S4c), and the second at 500 °C associated with PTFE decomposition. The DTA thermogram confirmed these results by showing an initial endothermic peak near 325 °C, corresponding to PTFE melting, followed by another peak related to PEO decomposition. A second major endothermic event appeared around 580 °C, aligned with PTFE decomposition, as indicated by the significant mass loss in the TGA curve [54]. Above 600 °C, the PTFE-SiO_2 NF mat retained less than 2% of its original weight, due to the thermally stable nature of SiO_2 NPs, which can withstand temperatures up to 850 °C [55]. In Fig. 3f, the DSC curves showed two distinct melting peaks at 63.3 °C and 340.13 °C, corresponding to PEO and PTFE phases, respectively [56]. In addition, Fig. S4d shows no mass loss at 380 °C, the decomposition temperature of PEO, indicating that the PEO component was completely removed during the sintering process of the PEO/PTFE-SiO_2 NF mat.

Figure S4e and 3g show the FTIR spectra of PTFE and the PTFE-SiO_2 NF mats before and after sintering, respectively. Fig. S4e presents the FTIR spectra of the PEO/PTFE and PEO/PTFE-SiO_2 NF mats (*i.e.*, before sintering); the peaks originating from C – H, C – O, C – O – C, and C – F vibrations indicate the coexistence of PEO and PTFE. The broad signal at 2877 cm^{-1} was attributed to asymmetric C – H stretching in CH_2, while the weak signals at 1342 and 1467 cm^{-1} originated from C – H bending. The bands at 1100 and 1060.8 cm^{-1} were associated with C – O – C stretching [57, 58], and the bands at 962 and 842 cm^{-1} corresponded to symmetric C – O stretching [59]. Compared to the relatively weak signals of PEO, PTFE exhibited prominent signals. The significant bands at 1203 and 1147 cm^{-1} were assigned to asymmetric and symmetric C – F stretching in CF_2, and the bands at 638 and 553 cm^{-1} were attributed to chain wagging and bending of the C – F bond [59–63]. After sintering at 380 °C (Fig. 3g), the peaks related to C – H and C – O vibrations disappeared, confirming the removal of PEO, while the C – F vibration bands remained, indicating the retention of PTFE in the mat. In addition, the bands at 1113 and 794 cm^{-1}, corresponding to asymmetric and symmetric Si – O – Si stretching (Fig. S4f) [61, 63], confirmed the preservation of SiO_2 after sintering [60, 64].

In addition, incorporating SiO_2 NPs, which contain the nonpolar Si – O – Si chains, also influenced the water contact angle (WCA) (Fig. 3h). This can be explained using the Cassie-Baxter model, which states that increased surface roughness reduces the effective contact area between water droplets and the solid surface, leading to higher hydrophobicity [65, 66]. The SiO_2 NPs increased the surface roughness of the PTFE NFs by facilitating the formation of wrinkles and nano-scale protrusions on the fiber surface (cf. Fig. 2 and Fig. S5), which enables the trapping of more air pockets beneath the water droplets, reducing adhesion and increasing WCA. This added roughness enhances the Cassie–Baxter state, in which water droplets remain suspended with minimal solid contact, leading to improved hydrophobic properties. These combined effects hindered the interaction of the PTFE-SiO_2 NF mats with polar molecules, enhancing their hydrophobic properties [23, 67].

Fig. 3h shows the WCAs of the PTFE-SiO_2 NF mats with varying SiO_2 concentrations. The pristine PTFE NF mat exhibited a super-hydrophobicity with a WCA of 151°. At 4 and 7 wt% SiO_2, the WCA gradually increased to 152° and 156°, respectively, because of the increased surface porosity and roughness of the NFs (cf. Fig. 2b – e and Fig. S5). The enhanced surface porosity promoted air trapping on the NF surface, reducing the solid–liquid contact area and thereby increasing the WCA [68–70]. However, as the SiO_2 concentration increased to 10 wt%, the WCA decreased to 153°, due to possible SiO_2 NP agglomeration and uneven dispersion along the PTFE NF matrix, which led to an increase in fiber D_{avg}, thereby reducing porosity and surface roughness (Fig. S5) [71–73].

3.3 Energy-harvesting Performance and Applications of PTFE-SiO_2 Nanofiber-Based Triboelectric Nanogenerator

Figure 4a-b display the frequency (*f*)-dependent output voltage (*V*) and short-circuit current (*I*) signals of the pristine PTFE and PTFE-SiO_2 (7 wt%) NF TENGs. Figure S6a-b display the *V* and *I* signals for PTFE-SiO_2 NF (4 and 10 wt%) TENG, under a fixed applied force (*F*) of 20 N and a contact distance (*h*) of 5 mm. The overall *V* and *I* signals of the pristine PTFE and PTFE-SiO_2 NF TENGs increased with *f*. As *f* increased from 2 to 10 Hz, the maximum output voltage (V_{max}) of the pristine PTFE NF TENG gradually increased from 8.7 to 62.6 V with a corresponding increase in the maximum output current (I_{max}) from 0.98 to 6.74 μA.

While the energy-harvesting performance improved with increasing the SiO_2 concentration from 0 to 7 wt% (Fig. S6a and b), the V_{max} and I_{max} values of the PTFE-SiO_2 (10 wt%) NF TENG at 10 Hz decreased to 113 V and 12 μA, respectively. This degradation in performance can be attributed to the increased D_{avg} at the SiO_2 concentrations exceeding 7 wt% (Fig. 2b – e). The increase in D_{avg} and decrease in surface roughness reduced the effective electrification contact area between the PTFE-SiO_2 NF mat and Nylon 6/6 NF mat, lowering the efficiency of charge transfer and decreasing the device performance [74–76].

The incorporation of SiO_2 NPs into the PTFE NFs significantly improved the energy-harvesting performance of TENG. The V_{max} value of the PTFE-SiO_2 (7 wt%) NF

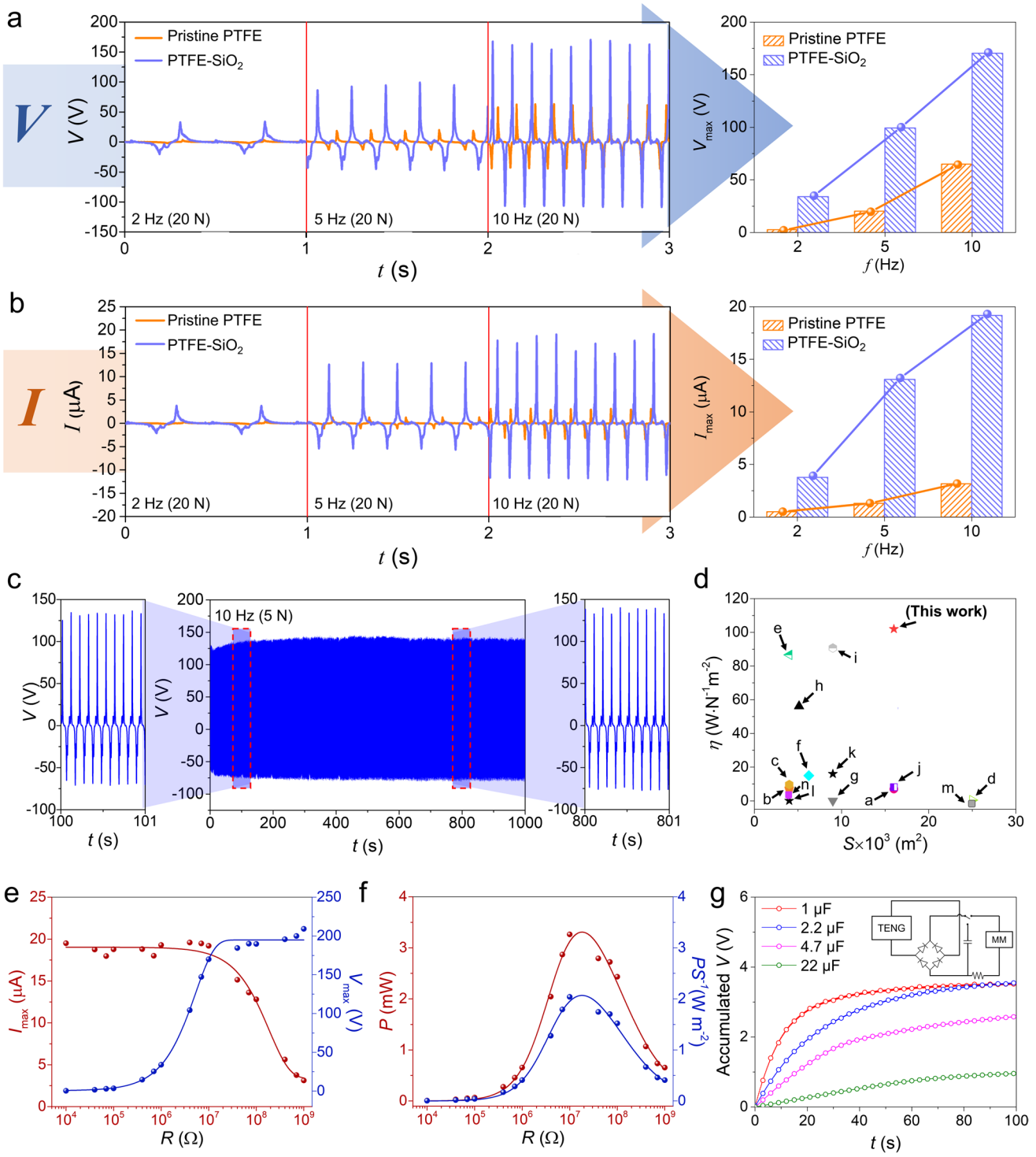


Fig. 4 Kinetic energy-harvesting performance of pristine PTFE and PTFE-SiO_2 (7 wt%) NF TENGs. **a** *V* signals of pristine PTFE and PTFE-SiO_2 (7 wt%) NF TENGs with *f* varying from 2 to 10 Hz at a fixed *F* of 20 N. **b** Corresponding *I* signals. **c** Energy-harvesting performance of PTFE-SiO_2 (7 wt%) NF TENG for 10,000 tapping cycles at a fixed *f* of 10 Hz and *F* of 5 N. **d** Comparison of energy-harvesting efficiency of the previously reported NF-based TENGs (cf. Table S1). Output signals of PTFE-SiO_2 NF TENG with an increase in *R* from 10 kΩ to 1 GΩ: **e** V_{max} and $I_{max,}$ and **f** *P* and *P*/*S*. **g** Charging curves using different capacitors

TENG increased from 34.1 to 170.4 V with increasing frequency, corresponding to an increase in I_{max} value from 3.8 to 19.2 μA. This enhancement in energy-harvesting performance was attributed to the enhanced surface potential (φ) of the PTFE-SiO_2 NF mat. The φ of the pristine PTFE NF mat with a work function (ϕ) of 5.66 eV was measured to be − 0.946 V. In contrast, the PTFE-SiO_2 NF mat with a ϕ value of 6.94 eV exhibited a more negative φ value of − 2.19 V (Fig. S7a and b). This indicated that the PTFE-SiO_2 NF mat possessed a higher negative surface charge than the pristine PTFE NF mat.

Surface charges on TENG often decay rapidly due to environmental factors, including temperature, humidity, pressure, and surface conductivity. However, if deep trap sites capture surface charges, they can form space charges that decay slowly, leading to more stable and sustained charge retention [77, 78]. PTFE–SiO_2 NF TENG maintained stable output performance across a broad temperature range (Fig. S8a). The output voltage increased with temperature and reached a maximum of 86 V at 60 °C, which can be attributed to thermally enhanced interfacial electron transfer [79, 80]. Even at 80 °C, the TENG demonstrated an output of ~ 79 V, indicating its high-temperature tolerance with negligible performance degradation.

In contrast, humidity had a more pronounced influence on the TENG (Fig. S8b). As the relative humidity increased from 20 to 80%, the output voltage decreased from 78 to 57 V, corresponding to a reduction of ~ 26.9%. This degradation can be attributed to the adsorption of water molecules on the fiber surface, which increases surface conductivity and facilitates partial charge dissipation [81, 82]. Nevertheless, the TENG retained sufficiently high output under humid conditions, underscoring its robustness and practical applicability.

SiO_2 NPs are known for their high surface charge density due to abundant trap sites, such as dangling bonds, which effectively capture and hold charges [83, 84]. Incorporating SiO_2 NPs into the matrix of PTFE NF enhances the electric performance of the TENG through multiple mechanisms. The SiO_2 NPs decreased the PTFE fiber diameter (Fig. 2) and increased the surface roughness of the PTFE NFs (Fig. S5), thereby expanding the effective contact area and enhancing charge transfer efficiency during contact-separation cycles. In addition, the high dielectric constant of SiO_2 promotes efficient charge trapping and storage, thereby reducing charge leakage. The deep trap sites on SiO_2 surfaces facilitate the formation of stable space charges, slowing charge decay and maintaining a higher charge density [73].

To further explore the potential distribution (ψ) of the pristine PTFE and PTFE-SiO_2 NF TENGs during contact-separation cycles, computational simulations were conducted using COMSOL Multiphysics, incorporating the actual material properties and dimensions of the TENGs (Fig. S9a-b). The results showed that the PTFE-SiO_2 NF TENG could produce a higher potential than the pristine PTFE NF TENG, which agrees well with the experimental findings (cf. Fig. 4a and b).

The energy-harvesting performance of the PTFE-SiO_2 NF TENG depended on f and h, whereas being independent of F, as shown in Fig. S10a and b. These results can be attributed to the interrelation among the V, h, and number of triboelectric charges (Q) [85–87]. When two triboelectric materials with opposite triboelectric properties within a TENG come into physical contact, surface charges are exchanged, contributing to the formation of an electric field between the two electrodes of the TENG. The V during the contact can be expressed by Eq. S1 and simplified as Eq. S2 by incorporating the effective thickness constant (d_0). By integrating Eq. S2 with Ohm's law, Eq. S3 is obtained and subsequently solved under the boundary condition $Q\ (t=0)=0$ as follows:

$$Q(t)=\left[\int_0^t \frac{\sigma h(\tau)}{R\varepsilon_a} e^{\int_0^\tau \frac{d_0+h(z)}{RS\varepsilon_a}dz} d\tau\right] e^{-\int_0^t \frac{d_0+h(\tau)}{RS\varepsilon_a}d\tau} \tag{2}$$

where σ, R, ε_a, and S are the charge density, the external resistance, the permittivity of air, and the area, respectively. Using Eq. 2, $I(t)$ and $V(t)$ can be then expressed as:

$$I(t)=\frac{dQ}{dt} = \frac{\sigma h(t)}{R\varepsilon_a} - \frac{d_0+h(t)}{RS\varepsilon_a} e^{-\int_0^t \frac{d_0+h(\tau)}{RS\varepsilon_a}d\tau} \cdot \int_0^t \frac{\sigma h(\tau)}{R\varepsilon_a} e^{\int_0^\tau \frac{d_0+h(z)}{RS\varepsilon_a}dz} d\tau \tag{3}$$

and

$$V(t)=I(t)\cdot R = \frac{\sigma h(t)}{\varepsilon_a} - \frac{d_0+h(t)}{S\varepsilon_a} e^{-\int_0^t \frac{d_0+h(\tau)}{RS\varepsilon_a}d\tau} \cdot \int_0^t \frac{\sigma h(\tau)}{R\varepsilon_a} e^{\int_0^\tau \frac{d_0+h(z)}{RS\varepsilon_a}dz} d\tau. \tag{4}$$

When the top electrode begins to separate from the dielectric layer, $h(t)$ is expressed as $(A_0/2)\cdot[1-\cos(\omega t)]$, where A_0 and ω are the amplitude and the angular velocity, respectively. The numerical results of $Q(t)$, $I(t)$, and $V(t)$ during this motion were obtained from Eqs. 2–4 by utilizing the parameters listed in Table 2.

Figure S11a, b show the numerical results of $Q(t)$, $I(t)$, and $V(t)$ as functions of f. The maximum Q (Q_{max}) value decreases significantly with increasing f, due to the shorter contact time at higher f [88]. In contrast, both the I_{max} and V_{max} values gradually increase with increasing f because of the improved change rate in $Q(t)$ (Fig. S11b). This finding

Table 2 Parameters for numerical calculations of energy-harvesting performance

Parameter	Value
Surface triboelectric charge density (σ)	15 [$\mu C \cdot m^{-2}$]
External resistance (R)	100 [MΩ]
Area (S)	16 [cm^2]
Effective thickness constant (d_0)	50 [μm]
Permittivity of air (ε_a)	$8.85 \cdot 10^{-12}$ [$F \cdot m^{-1}$]
Amplitude of the motion (A_0)	5 [mm]
Angular velocity (ω)	$2\pi f$ [$rad \cdot s^{-1}$]

aligns with the experimental results for I_{max} and V_{max} (see Fig. S12a).

Furthermore, h, which depends on A_0, influenced the magnitude of $Q(t)$, $I(t)$, and $V(t)$, as shown in Fig. S11c. As h increases, the velocity (v) of the motion, given by $v = h/f$, also increases because of the fixed f. The increase in v reduced the number of triboelectric charges returning to their original state, resulting in an increase in Q_{max} (Fig. S12b) [86]. In addition, the increase in $Q(t)$ with increasing h contributed to the improvement of both I_{max} and V_{max}, which was similar to the trends observed in the experimental results (Fig. S12c).

The PTFE-SiO_2 (7 wt%) NF TENG exhibited stable and durable energy-harvesting performance even under repeated operational conditions. Figure 4c presents the cyclic energy-harvesting performance of the PTFE-SiO_2 NF TENG for 10,000 tapping cycles under a fixed F and f of 5 N and 10 Hz, respectively. While minor fluctuations in the output signal were observed during the initial cycles, the TENG demonstrated outstanding stability and reliability while maintaining a consistent V of 130 V. Furthermore, the surface charge distribution of the PTFE-SiO_2 NF TENG was also evaluated through current signal analysis under a fixed F and f of 10 N and 5 Hz, respectively (Fig. S13a). The positive and negative surface charges, denoted as q_{pos} and q_{neg}, respectively, were quantified by integrating the current over time. The symmetrical charge distribution of q_{pos} and q_{neg}, measured at 7.848×10^{-8} and 7.794×10^{-8} C, respectively, confirmed the TENG's stable and reliable performance.

The energy-harvesting performance of the PTFE-SiO_2 NF TENG was compared with that of previously reported NF-based TENGs based on their efficiency (η; $\eta = P \cdot F^{-1} \cdot S^{-1}$, where P, F, and S represent the maximum output power, applied force, and sample size, respectively) (Fig. 4d and Table S1). NF-based TENGs for which P, F, and S were unavailable were excluded from this comparison. The PTFE-SiO_2 NF TENG achieved an η value of 102 $mW \cdot N^{-1} \cdot m^{-2}$, outperforming previously reported NF-based TENGs that exhibited η values below 90.75 $mW \cdot N^{-1} \cdot m^{-2}$. The outstanding triboelectric performance of the PTFE–SiO_2 NF TENG is attributed to the synergistic contributions of three key factors: (i) the high electron affinity of PTFE, (ii) the charge-trapping capability imparted by SiO_2 NPs, and (iii) the morphology optimization achieved through electrospinning. Consequently, these effects promote efficient charge generation and long-term retention, thereby distinguishing the PTFE–SiO_2 NF TENG from previously reported materials. The output power of the PTFE-SiO_2 NF TENG was further investigated by applying and increasing the external load (R) from 10 kΩ to 1 GΩ (Fig. 4e and f). With increasing R, the V_{max} value gradually increased to 209.3 V, corresponding to a decrease in I_{max} from 19.5 to 3.13 μA. The maximum P and output power density (P/S) at $R = 10$ MΩ were recorded as 3.26 mW and 2.04 W/m^2, respectively. The rectified electrical energy generated from the PTFE-SiO_2 NF TENG could be stored using different capacitances of 1, 2.2, 4.7, and 22 μF with a full-wave bridge rectifier circuit (Fig. 4g). The results demonstrated that the PTFE-SiO_2 NF TENG could charge all capacitors within 100 s under F of 20 N and f of 5 Hz.

To demonstrate the practical applications of the PTFE-SiO_2 NF TENG, its energy-harvesting performance in single-electrode mode was explored under conditions involving contact with human skin and water droplets, which acted as a tribopositive material. Figure 5a illustrates the underlying mechanism of contact electrification between human skin and PTFE-SiO_2 NF mat through friction. The energy-harvesting performance from human skin contact was examined by varying f (Fig. 5b and Fig. S13b and c). As f increased from 1 to 5 Hz at a fixed h of 5 cm, the averaged V value increased from 8.7 to 15.5 V, corresponding to an increase in I value from 1.14 to 2.3 μA. Under random applied force, the PTFE-SiO_2 NF TENG exhibited enhanced energy-harvesting performance, achieving V_{max} and I_{max} values of 65.3 V and 6.57 μA, respectively (Fig. 5c). Furthermore, the PTFE-SiO_2 NF TENG revealed a rapid response time of 10 ms and high stability, as indicated by the symmetrical magnitude of q_{pos} and q_{neg} values (Fig. S13d).

The energy-harvesting performance of the PTFE-SiO_2 NF TENG for water droplets was evaluated by varying the height (d) of the contact-sliding water droplets on the TENG (Fig. 5d). Before contact with the PTFE-SiO_2 NF mat, water droplets already contain triboelectric charges on their surface due to friction with air/pipes [89–91]. Due to contact electrification and frictional charge transfer, the water droplets acquire triboelectric charges when falling through air or detaching from tubes/pipes, or, in this case, a metallic needle tip. As a droplet forms, it remains in contact with the needle, exchanging electrons based on the needle material's triboelectric properties. Upon detachment, friction between the water and the needle surface induces charge imbalance, often leaving the droplet electrically charged [92, 93].

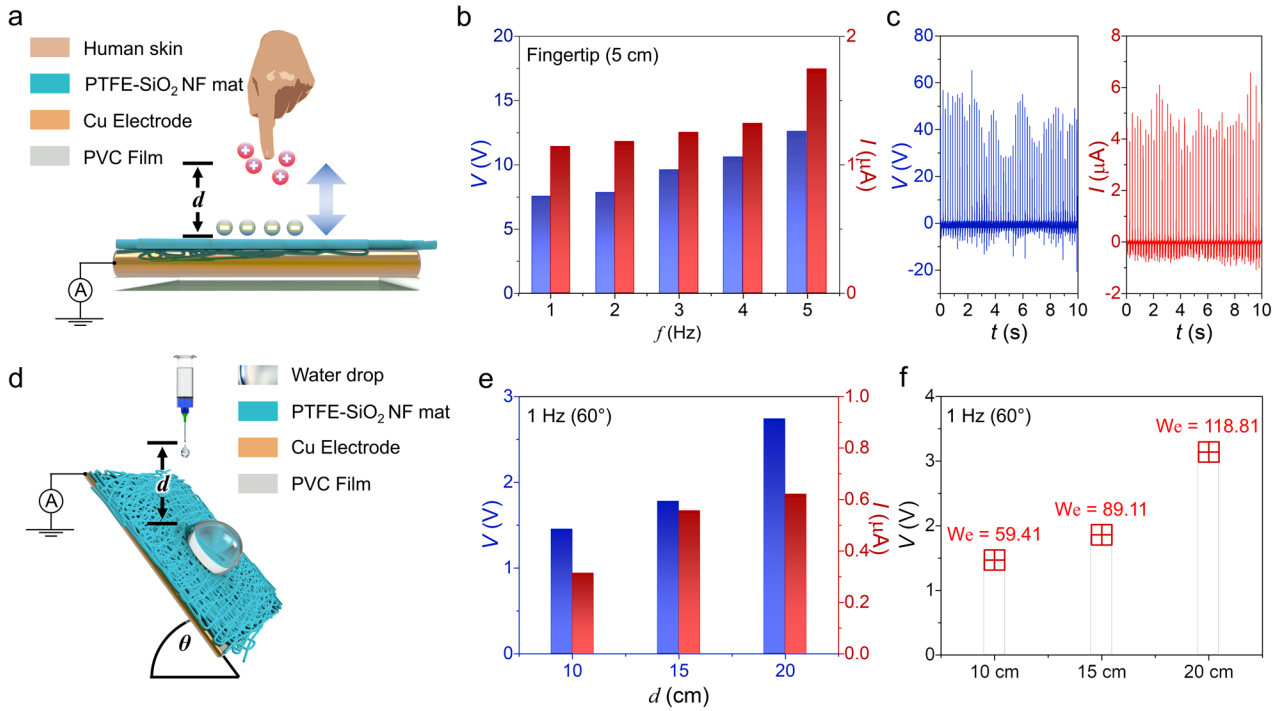


Fig. 5 **a** Energy harvesting of the single-mode PTFE-SiO_2 NF TENG with human skin. **b** Averaged V and I signals of PTFE-SiO_2 NF TENG when contacted with a fingertip at varying f under fixed F and h of 0.5 N and 5 cm, respectively. **c** V and I signals of PTFE-SiO_2 NF TENG upon direct contact with human skin. **d** Energy-harvesting of the single-mode PTFE-SiO_2 NF TENG with water droplets. **e** Averaged V and I signals of PTFE-SiO_2 NF TENG in response to water droplets with varying h at a fixed impact angle and f of 60° and 1 Hz, respectively. **f** Averaged V and W_e values of PTFE-SiO_2 NF TENG as a function of d

When a water droplet contacts the PTFE-SiO_2 NF mat (Stage II in Fig. S14a), an electric potential is created between the Cu electrode and the ground. The electrons are transferred from the ground to the electrode to balance the potential difference until reaching equilibrium (Stage III in Fig. S14a). When the water droplet slides off the PTFE-SiO_2 NF mat, it creates a negative electric potential; therefore, electrons must flow from the Cu electrode to the ground until a new equilibrium is reached (Stage IV in Fig. S14a). If the water flow is maintained, a continuous output can be generated. The hydrophobicity of PTFE-SiO_2 NF TENG allows water droplets to roll down the tilted TENG surface, facilitating charge generation through repeated contact and separation between the PTFE-SiO_2 NF mat surface and the water droplet [90, 94, 95].

Figure 5e shows the average V and I values of the PTFE-SiO_2 NF TENG with varying d. As d increased from 10 to 20 cm (Fig. S14b and c), the averaged V value increased from 1.5 to 2.7 V, accompanied by an increase in the averaged I value from 0.28 to 0.61 μA. This improvement was attributed to the larger contact area, which resulted from the higher impact of water droplets as d increased [96–98]. The dynamics of impacting water droplets on the hydrophobic surface were estimated using the Weber number (W_e), a dimensionless parameter that represents the ratio of kinetic energy to surface energy [98–100], as follows [101]:

$$W_e = \frac{\rho v_0 R_0}{\sigma_s}, \tag{5}$$

where ρ is the fluid density, v_0 is the fluid velocity, R_0 is the drop radius, and σ_S is the liquid surface tension. The previously reported studies demonstrated that the maximum spreading area of a water droplet is proportional to W_e, owing to the increased impact inertia at higher W_e values [102–104]. Therefore, the water droplets with a high W_e value contribute to an enlarged contact area, thereby enhancing energy-harvesting performance. In our experiment, for water droplets with a volume of 45 μL, the values of ρ, σ_s, and R_0 were 1,000 kg·m^{-3}, 0.073 N·m^{-1}, and 2.2 mm, respectively. As v_0 increased from 1.4 to 1.98 m·s^{-1} with a change in d from 10 to 20 cm, the corresponding W_e increased from 59.4 to 118.8. This increase in W_e improved the electrical output of the PTFE-SiO_2 NF TENG (Fig. 5f), which was attributed to the enhancement in the transferred charges between the water droplets and the PTFE-SiO_2 NF surface [105, 106].

In addition, the highly stable energy-harvesting performance of the PTFE-SiO_2 NF TENG with water droplets was also validated by the symmetric distributions of q_{pos} and q_{neg} (Fig. S14d). To further demonstrate the practical energy-harvesting capability of the PTFE-SiO_2 NF TENG for raindrops, water was manually poured onto the TENG without controlling the flow rate. As shown in Fig. S14e, the V signal remained at approximately 0 V prior to the introduction of water. However, upon contact with the water droplets, a change in the V signal was immediately observed, reaching a maximum V of 4.3 V.

To further explore the feasibility of PTFE-SiO_2 NF TENG as power sources for fabric-based wearable devices, the deformability and durability of the PTFE-SiO_2 NF TENG fabricated using a conductive cloth (Fig. 6a) were evaluated through various mechanical deformations, including folding, twisting, and self-cleaning tests. The PTFE-SiO_2 NF TENG, with a size of 40 mm × 25 mm, demonstrated high flexibility, maintaining its structure stability under folding and twisting (Fig. 6b), which can easily be attached to any cloth (Fig. 6c). In addition, the superhydrophobic nature of the PTFE-SiO_2 NF TENG provided resistance to wetting from various liquids, including coffee, cola, tea, milk, and a 1 M salt solution (Fig. 6d). This feature facilitated effective self-cleaning, as contaminants (*i.e.*, soil) could be easily removed when water droplets rolled off the TENG surface. (Fig. 6e and Movie S1). The self-cleaning stability of the PTFE-SiO_2 NF mat under long-term environmental exposure was evaluated

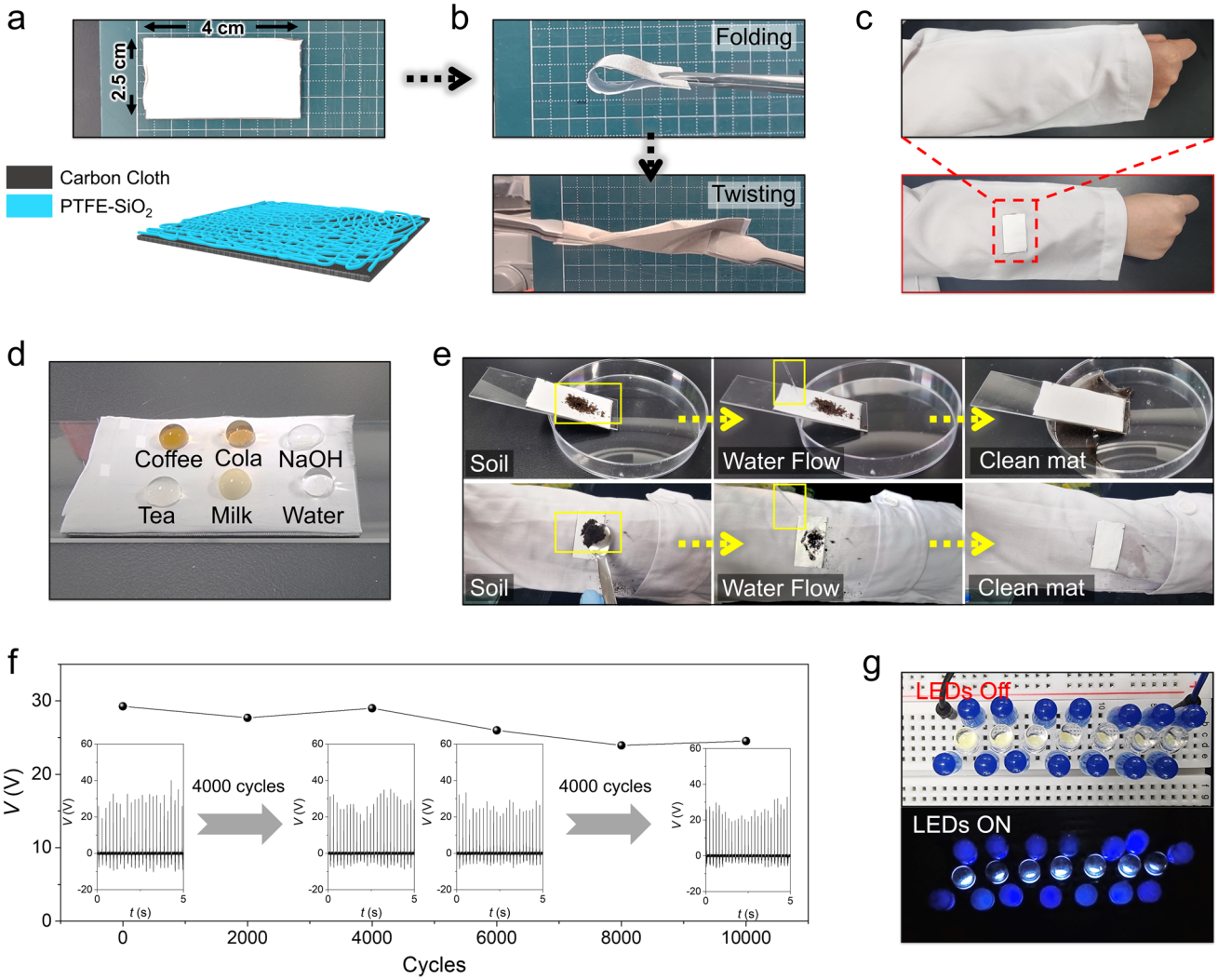


Fig. 6 **a** Photo and illustration of the wearable PTFE-SiO_2 NF TENG with a conductive cloth as an electrode. Photos of **b** the deformability of wearable PTFE-SiO_2 NF TENG during folding and twisting, and **c** the TENG attached on the cloth. **d** Hydrophobicity of the TENG against different liquids and **e** the corresponding self-cleaning performance. **f** V signal of the wearable PTFE-SiO_2 NF TENG during finger tapping after undergoing folding deformation for up to 10,000 cycles. **g** LED operation of the single-electrode-based wearable PTFE-SiO_2 NF TENG

using the same samples shown in Fig. 6d-e. Even after several months of ambient storage without protective conditions, the mat retained high hydrophobicity and self-cleaning capability (Fig. S15a–d), exhibiting only a slight reduction in WCA, from 156° to 152° (Fig. S15e-f). FTIR analysis further confirmed that the characteristic $–CF_2$ and $–CF_3$ peaks remained unchanged after prolonged exposure (Fig. S16), highlighting the inherent chemical stability of PTFE [107]. The sustained self-cleaning performance originated from the intrinsically low surface energy of PTFE, associated with $–CF_2$ and $–CF_3$ groups, and the hierarchical roughness of the nanofibrous structure [107, 108]. This synergistic effect preserves superhydrophobicity and ensures efficient contaminant removal over extended periods [109]. As shown in Fig. S17, the mat maintained robust self-cleaning durability even after 20 washing cycles, exhibiting only a slight reduction in WCA from 156° to 152°, while still retaining a distinctly superhydrophobic property.

Furthermore, the superhydrophobic durability of PTFE–SiO_2 NF mat was evaluated by comparing the WCA after three different mechanical resistance tests: linear abrasion, adhesive tape-peeling, and scratch testing. As shown in Fig. S18a, the mat withstood 30 abrasion cycles, and the SEM analysis (Fig. S18b) revealed that the fibrous structure was preserved, with only minor surface fiber breakage observed on the surface. In the adhesive tape-peeling test, the fibers remained firmly attached to the substrate, showing no evidence of detachment, thereby indicating strong interfacial stability. In the scratch test, while the applied force cut the surface fibers, the hierarchical fibrous network remained intact and continued to exhibit hydrophobicity. Notably, the WCA results showed that the PTFE–SiO_2 NF mat maintained superhydrophobic properties even after all mechanical resistance tests (Fig. S18c), thereby confirming both the structural stability and the mechanical robustness of the mat.

The stability of the PTFE-SiO_2 NF-based TENG performance under mechanical deformation was evaluated. During bending (Fig.S19a), the TENG exhibited a stable output voltage of approximately 35 V, with peaks up to 40 V over 200 cycles. Under twisting, the output decreased to ~ 12 V, due to the reduced effective contact area (Fig. S19c). As shown in Fig. S19b and d, SEM analysis verified no significant changes in fiber morphology after testing, demonstrating that the PTFE-SiO_2 NF mat preserves both structural integrity and reliable performance under repeated mechanical deformation. The robustness of the PTFE-SiO_2 NF TENG was further demonstrated through a cyclic bending test, where the output V signal remained stable, retaining over 85% of its initial value even after 10,000 folding cycles at 5 Hz (Fig. 6f and Movie S2). Corresponding SEM images (Fig. S20) revealed that the fibrous morphology was preserved after cycling, with only minor surface defects. These results indicate that the structural stability of PTFE-SiO_2 NFs is maintained upon the incorporation of SiO_2 NPs, thereby sustaining its triboelectric output performance**.**

Finally, after undergoing multiple deformations, the PTFE-SiO_2 NF TENG was able to power 21 light-emitting diodes (LEDs) under a specific contact condition ($F < 2$ N, $f = 3$ Hz), demonstrating its durability and stable performance under external mechanical stresses (Fig. 6g and Movie S3).

4 Conclusion

In this study, a self-cleanable, wearable, and superhydrophobic PTFE-SiO_2 nanofiber (NF) mat was fabricated via a facile electrospinning method and demonstrated as a highly tribonegative material for high-performance triboelectric nanogenerators (TENGs). The one-step aqueous electrospinning approach not only enables precise control of fiber morphology but also offers an eco-friendly fabrication route. The incorporation of SiO_2 nanoparticles (NPs) significantly increases surface roughness and charge-trapping sites, leading to improved triboelectric output compared with conventional PTFE-based systems. As a result, the electrical output of the PTFE-SiO_2 NF-based TENG improved nearly threefold, achieving 170 V under a tapping force of 20 N at 10 Hz, with a corresponding maximum power density of 2.04 $W{\cdot}m^{-2}$. The strong tribonegativity of the composite enabled efficient energy harvesting in single-electrode mode, utilizing both human motion and water droplets as mechanical stimuli. These results highlight the composite's promising potential for self-powered wearable electronics and eco-friendly energy-harvesting systems.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42765-025-00633-7.

Acknowledgements This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean Government (MSIT) (No. RS-2023-00211303) and by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (RS-2023-KI002684, HRD Program for Industrial Innovation). This research was also funded and conducted under the Competency Development Program for Industry Specialists of the Korean Ministry of Trade, Industry and Energy (MOTIE), operated by Korea Institute for Advancement of Technology (KIAT) [No. P0023704, Semiconductor-Track Graduate School (SKKU)].

Funding National Research Foundation of Korea (NRF), RS-2023-00211303, Seongpil An, Korea Institute for Advancement of Technology, RS-2023-KI002684, Seongpil An, P0023704, Seongpil An.

Declarations

Conflict of interest The authors declare no conflict of interest. They have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Kang M, Yum HY, Kim HT, Park BJ, Cho DS, Choi Y, Kim HJ, Cho Y, Kim YJ, Lee DM, Lee DG, Song HC, Nam SH, Lee JH, Choi BO, Kim SW. Self-powered electrical bandage based on body-coupled energy harvesting. *Adv Mater*. **2024**;*36*:2402491.
2. Fan X, Ke T, Gu H. Multifunctional, ultra-tough organohydrogel E-skin reinforced by hierarchical goatskin fibers skeleton for energy harvesting and self-powered monitoring. *Adv Funct Mater*. **2023**;*33*:2304015.
3. Veenuttranon K, Kaewpradub K, Jeerapan I. Screen-printable functional nanomaterials for flexible and wearable single-enzyme-based energy-harvesting and self-powered biosensing devices. *Nano-Micro Lett*. **2023**;*15*:85.
4. Khumngern S, Jeerapan I. Synergistic convergence of materials and enzymes for biosensing and self-sustaining energy devices towards on-body health monitoring. *Commun Mater*. **2024**;*5*:135.
5. Song KY, Kim SW, Nguyen DC, Park JY, Luu TT, Choi D, Baik JM, An S. Recent progress on nature-derived biomaterials for eco-friendly triboelectric nanogenerators. *EcoMat*. **2023**;*5*:e12357.
6. Park D, Hong JH, Choi D, Kim D, Jung WH, Yoon SS, Kim KH, An S. Biocompatible and mechanically-reinforced tribopositive nanofiber mat for wearable and antifungal human kinetic-energy harvester based on wood-derived natural product. *Nano Energy*. **2022**;*96*:107091.
7. Cao X, Xiong Y, Sun J, Xie X, Sun Q, Wang ZL. Multidiscipline applications of triboelectric nanogenerators for the intelligent era of internet of things. *Nano Micro Lett*. **2023**;*15*:14.
8. Jo H, Park D, Joo M, Choi D, Kang J, Ha JM, Kim KH, An S. Performance-enhanced eco-friendly triboelectric nanogenerators via wettability manipulation of lignin. *EcoMat*. **2023**;*5*:e12413.
9. Lu P, Liao X, Guo X, Cai C, Liu Y, Chi M, Du G, Wei Z, Meng X, Nie S. Gel-based triboelectric nanogenerators for flexible sensing: principles, properties, and applications. *Nano Micro Lett*. **2024**;*16*:206.
10. Hasan MAM, Zhu W, Bowen CR, Wang ZL. Triboelectric nanogenerators for wind energy harvesting. *Nat Rev Electr Eng*. **2024**;*1*:453.
11. Qiu H, Wang H, Xu L, Zheng M, Wang ZL. Brownian motor inspired monodirectional continuous spinning triboelectric nanogenerators for extracting energy from irregular gentle water waves. *Energy Environ Sci*. **2023**;*16*:473.
12. Han J, Li J, Zhang X, Zhao L, Wang C. Enhancing the performance of triboelectric nanogenerators via chitosan films surface modification. *Chem Eng J*. **2024**;*489*:151493.
13. Meng H, Zhang J, Zhu R, Wang J, Ge Y, Liu H, Feng C, Zhou Z, Meng Y, Huang Z. Elevating outputs of droplet triboelectric nanogenerators through strategic surface molecular engineering. *ACS Energy Lett*. **2024**;*9*:2670.
14. Li X, Xu L, Lin P, Yang X, Wang H, Qin H, Wang ZL. Three-dimensional chiral networks of triboelectric nanogenerators inspired by metamaterial's structure. *Energy Environ Sci*. **2023**;*16*:3040.
15. Yang C, Wang Y, Wang Y, Zhao Z, Zhang L, Chen H. Highly stretchable PTFE particle enhanced triboelectric nanogenerators for droplet energy harvestings. *Nano Energy*. **2023**;*118*:109000.
16. Gao Y, Liu J, Zhou L, He L, Liu D, Yang P, Jin B, Wang ZL. Achieving high-performance triboelectric nanogenerators simultaneously with high-voltage and high-charge energy cycle. *Energy Environ Sci*. **2024**;*17*:8734.
17. Jo E, Kim H, Park YJ, Park H, Lee M. Characterization of corona discharge treatment on PTFE surface for TENG applications. *J Electrostat*. **2023**;*126*:103865.
18. Ahmed AA, Qahtan TF, Afzal N, Rashid M, Thalluri LN, Mohamed Ali MS. Low-pressure air plasma-treated polytetrafluoroethylene surface for efficient triboelectric nanogenerators. *Mater Today Sustain*. **2023**;*21*:100330.
19. Li J, Zhong Q, Yao Y, Bi S, Zhou T, Guo X, Wu M, Feng T, Xiang R. Electrochemical performance and thermal stability of the electrospun PTFE nanofiber separator for lithium-ion batteries. *J Appl Polym Sci*. **2018**;*135*:46508.
20. Hwang CK, Lee KA, Lee J, Kim Y, Ahn H, Hwang W, Ju BK, Kim JY, Yeo SY, Choi J, Sung YE, Kim ID, Yoon KR. Perpendicularly stacked array of PTFE nanofibers as a reinforcement for highly durable composite membrane in proton exchange membrane fuel cells. *Nano Energy*. **2022**;*101*:107581.
21. Zhao P, Soin N, Prashanthi K, Chen J, Dong S, Zhou E, Zhu Z, Narasimulu AA, Montemagno CD, Yu L, Luo J. Emulsion electrospinning of polytetrafluoroethylene (PTFE) nanofibrous membranes for high-performance triboelectric nanogenerators. *ACS Appl Mater Interfaces*. **2018**;*10*:5880.
22. Lin S, Cheng Y, Mo X, Chen S, Xu Z, Zhou B, Zhou H, Hu B, Zhou J. Electrospun polytetrafluoroethylene nanofibrous membrane for high-performance self-powered sensors. *Nanoscale Res Lett*. **2019**;*14*:251.
23. Zou F, Li G, Wang X, Yarin AL. Dynamic hydrophobicity of superhydrophobic PTFE–SiO_2 electrospun fibrous membranes. *J Membr Sci*. **2021**;*619*:118810.
24. Sun C, Zu G, Wei Y, Song X, Yang X. Flexible triboelectric nanogenerator based on electrospun poly(vinylidene fluoride) with MoS_2/Carbon nanotube composite nanofibers. *Langmuir*. **2022**;*28*:1479.
25. Kumar S, Jha RK, Thakur B, Biswas T, Anad JK, Soren C, Banswar D, Singh S, Singh S, Sinha-Ray S, Goswani A. Performance enhancement of triboelectric nanogenerators and exploration of tactile sensing using an electrospun PAN-MWCNT layer through interface manipulation. *J Mater Chem A*. **2025**;*13*:776.
26. Kochakkadan ST, Duque M, Murillo G, Nirwan VP, Fahmi A. Sustainable electrospun hybrid nanofibers for triboelectric nanogenerators. *Small*. **2025**;*21*:2410271.
27. Venkatesan M, Chandrasekar J, Hsu YC, Sun TW, Li PY, King XT, Chung MA, Chung RJ, Lee WY, Ye Z, Lin JH, Kuo CC. Rationally improved surface charge density of triboelectric nanogenerators with TiO_2-MXene/Polystyrene nanofiber charge trapping layer for biomechanical sensing and wound healing application. *Adv Sci*. **2024**;*11*:2404019.
28. Liu Z, Muhammad M, Cheng L, Xie E, Han W. Improved output performance of triboelectric nanogenerators based on polydimethylsiloxane composites by the capacitive effect of embedded carbon nanotubes. *Appl Phys Lett*. **2020**;*117*:143903.
29. Liu Z, Wang K, Jiang X, Javed MS, Han W. Overcoming current leaks in CNT/PDMS triboelectric composites by wrapping CNTs with TiO_2 insulation layer. *Appl Phys Lett*. **2022**;*121*:113902.
30. Samsudin EM, Hamid SBA. Effect of band gap engineering in anionic-doped TiO_2 photocatalyst. *Appl Surf Sci*. **2017**;*391*:326.
31. Green AE. Hypo-elasticity and plasticity. *Proc R Soc Lond A*. **1956**;*234*:1196.
32. Zhang W, Staszel C, Yarin AL, Shim E, Pourdeyhimi B. Point-bonded polymer nonwovens and their rupture in stretching. *Polymer*. **2018**;*146*:209.
33. An S, Yarin AL. Mechanical behavior of sintered submicron glass fiber mats. *Int J Mech Sci*. **2020**;*170*:105354.
34. Yarin AL, Pourdeyhimi B, Ramakrishna S. Fundamentals and applications of micro- and nanofibers. Cambridge: Cambridge University Press; 2014.
35. Xu C, Zi Y, Wang A, Zou H, Dai Y, He X, Wang P, Wang Y, Feng P, Li D, Wang ZL. On the electron-transfer mechanism in the contact-electrification effect. *Adv Mater*. **2018**;*30*:1706790.

36. Wang J, Wen Z, Zi Y, Lin L, Wu C, Guo H, Xi Y, Xu Y, Wang ZL. Self-powered electrochemical synthesis of polypyrrole from the pulsed output of a triboelectric nanogenerator as a sustainable energy system. *Adv Funct Mater*. **2016**;*26*:3542.
37. Wang J, Wen Z, Zi Y, Zhou P, Lin J, Guo H, Xu Y, Wang ZL. All-plastic-materials based self-charging power system composed of triboelectric nanogenerators and supercapacitors. *Adv Funct Mater*. **2016**;*26*:1070.
38. An S, Sankaran A, Yarin AL. Natural biopolymer-based triboelectric nanogenerators via fast, facile, scalable solution blowing. *ACS Appl Mater Interfaces*. **2018**;*10*:37749.
39. Kim WG, Kim DW, Tcho LW, Kim JK, Kim MS, Choi YK. Triboelectric nanogenerator: structure, mechanism, and applications. *ACS Nano*. **2021**;*15*:258.
40. Alam SN, Shrivastava P, Shukla U, Garg K, Edara AC, Sahoo N. An introduction to triboelectric nanogenerators. *Nano Struct Nano Objects*. **2023**;*34*:100980.
41. Yu Y, Li H, Zhang X, Gao Q, Yang B, Wang ZL, Cheng T. Substantially boosting performance of triboelectric nanogenerators via a triboelectricification enhancement effect. *Joule*. **2024**;*8*:1855.
42. Diaz AF, Felix-Navarro RM. A semi-quantitative tribo-electric series for polymeric materials: the influence of chemical structure and properties. *J Electrostat*. **2004**;*62*:277.
43. Wu C, Wang A, Ding W, Guo H, Wang ZL. Triboelectric nanogenerator: a foundation of the energy for the new era. *Adv Energy Mater*. **2019**;*9*:1802906.
44. Pan S, Zhang Z. Triboelectric effect: a new perspective on electron transfer process. *J Appl Phys*. **2017**;*122*:144302.
45. Zou H, Zhang Y, Guo L, Wang P, He X, Dai G, Zheng H, Chen C, Wang AC, Xu C, Wang ZL. Quantifying the triboelectric series. *Nat Commun*. **2019**;*10*:1427.
46. Wang W, Zhang L, Wang H, Zhao Y, Cheng J, Meng J, Wang D, Liu Y. High-output single-electrode droplet triboelectric nanogenerators based on asymmetrical distribution electrostatic induction enhancement. *Small*. **2023**;*19*:2301568.
47. Su C, Li Y, Cao H, Lu C, Li Y, Chang J, Duan F. Novel PTFE hollow fiber membrane fabricated by emulsion electrospinning and sintering for membrane distillation. *J Membr Sci*. **2019**;*583*:200.
48. Lee MW, Yarin AL, An S, Yoon SS. Self-healing nanotextured vascular engineering materials. Cham: Springer; 2019.
49. Dong ZQ, Ma XH, Xu ZL, Gu ZY. Superhydrophobic modification of PVDF–SiO_2 electrospun nanofiber membranes for vacuum membrane distillation. RSC Adv. 2015;5:67962.
50. Liang YY, Ju JG, Deng NP, Zhou XH, Yan J, Kang WM, Cheng BW. Super-hydrophobic self-cleaning bead-like PTFE-SiO_2 nanofiber membranes for waterproof-breathable applications. *Appl Surf Sci*. **2018**;*442*:54.
51. Delgado AO, Rizzutto MA, Severin D, Seidl T, Neumann R, Trautmann C. Latent track radius of PTFE irradiated with high energy ion beam. *Nucl Instrum Methods Phys Res B*. **2012**;*273*:55.
52. Xu H, Jin W, Wang F, Liu G, Li C, Wang J, Zhu H, Guo Y. Formation and characterization of polytetrafluoroethylene nanofiber membranes for high-efficiency fine particulate filtration. *RSC Adv*. **2019**;*9*:13631.
53. Lebedev YA, Korolev YM, Polikarpov VM, Ignat'eva LN, Antipov EM. X-ray powder diffraction study of polytetrafluoroethylene. *Crystallogr Rep*. **2010**;*55*:609.
54. Chiu J. Thermal analysis of high polymers: differential thermal analysis and dynamic electrothermal analysis. *J Polym Sci Part C*. **1965**;*8*:27.
55. Sasse HE, Konig U. SiO diffusion during thermal decomposition of SiO_2. *J Appl Phys*. **1990**;*67*:6194.
56. Dhanumalayan E, Joshi GM. Performance properties and applications of polytetrafluoroethylene (PTFE)—a review. *Adv Compos Hybrid Mater*. **2018**;*1*:247.
57. Dhatarwal P, Sengwa RJ. Influence of solid polymer electrolyte preparation methods on the performance of (PEO–PMMA)-LiBF4 films for lithium-ion battery applications. *Polym Bull*. **2018**;*75*:5645.
58. Li X, Liu H, Liu X, Fang N, Wang X, Wu J. Synthesis of bi-phase dispersible core–shell FeAu@ZnO magneto-opto-fluorescent nanoparticles. *Sci Rep*. **2015**;*5*:16384.
59. Kumar KK, Ravi M, Pavani Y, Bhavani S, Sharma AK, Rao VVRN. Investigations on PEO/PVP/NaBr complexed polymer blend electrolytes for electrochemical cell applications. *J Membr Sci*. **2014**;*454*:200.
60. Li L, Zi FT, Zheng YF. The characterization of fluorocarbon films on NiTi alloy by magnetron sputtering. *Appl Surf Sci*. **2008**;*255*:432.
61. Norton MG, Jiang WB, Dickinson JT, Hipps KW. Pulsed laser ablation and deposition of fluorocarbon polymers. *Appl Surf Sci*. **1996**;*96–98*:617.
62. Zhang XF, Dong XL, Huang H, Lv B, Zhu XG, Lei JP, Ma S, Liu W, Zhang ZD. Synthesis, structure and magnetic properties of SiO_2-coated Fe nanocapsules. *Mater Sci Eng A*. **2007**;*454–455*:211.
63. Feng YS, Zhou SM, Li Y, Li CC, Zhang LD. Synthesis and characterization of tin oxide nanoparticles dispersed in monolithic mesoporous silica. *Solid State Sci*. **2003**;*5*:729.
64. Wang N, Wang H, Ren J, Gao G, Chen S, Zhao G, Yang Y, Wang J. Novel additive of PTFE-SiO_2 core–shell nanoparticles with superior water lubricating properties. *Mater Des*. **2020**;*195*:109069.
65. Lafuma A, Quéré D. Superhydrophobic states. *Nat Mater*. **2003**;*2*:457.
66. Cassie ABD, Baxter S. Wettability of porous surfaces. *Trans Faraday Soc*. **1944**;*40*:546.
67. Toh MJ, Oh PC, Chew TL, Ahmad AL. Antiwettability enhancement of PVDF-HFP membrane via superhydrophobic modification by SiO_2 nanoparticles. *C R Chim*. **2019**;*22*:369.
68. Li JF, Guo SJ, Xu ZZ, Li JG, Pan ZH, Du ZP, Cheng FQ. Preparation of omniphobic PVDF membranes with silica nanoparticles for treating coking wastewater using direct contact membrane distillation: electrostatic adsorption vs. chemical bonding. *J Membr Sci*. **2019**;*574*:349.
69. Tran PA, Webster TJ. Understanding the wetting properties of nanostructured selenium coatings: the role of nanostructured surface roughness and air-pocket formation. *Int J Nanomed*. **2013**;*8*:2001.
70. Cho Y, Park CH. Objective quantification of surface roughness parameters affecting superhydrophobicity. *RSC Adv*. **2020**;*10*:31251.
71. Salehabadi S, Seyfi J, Hejazi I, Davachi SM, Naeini AH, Khakbaz M. Nanosilica-decorated sponges for efficient oil/water separation: role of nanoparticles' type and concentration. *J Mater Sci*. **2017**;*52*:7017.
72. Avci MZ, Sarac AS. Transparent poly(methyl methacrylate-co-butyl acrylate) nanofibers. *J Appl Polym Sci*. **2013**;*130*:4264.
73. Li WJ, Xiang YY, Zhang W, Loos K, Pei YT. Ordered mesoporous SiO_2 nanoparticles as charge storage sites for enhanced triboelectric nanogenerators. *Nano Energy*. **2023**;*113*:108539.
74. Lee JH, Yu I, Hyun S, Kim JK, Jeong U. Remarkable increase in triboelectrification by enhancing the conformable contact and adhesion energy with a film-covered pillar structure. *Nano Energy*. **2017**;*34*:233.
75. Chung CK, Ke KH. High contact surface area enhanced Al/PDMS triboelectric nanogenerators using novel overlapped

microneedle arrays and its application to lighting and self-powered devices. *Appl Surf Sci*. **2020**;*508*:145310.
76. Choi GR, Park J, Ha JW, Kim WD, Lim H. Superamphiphobic web of PTFEMA fibers via simple electrospinning without functionalization. *Macromol Mater Eng*. **2010**;*295*:995.
77. Wang J, Li G, Xu S, Wu H, Fu S, Shan C, He W, Zhao Q, Li K, Hu C. Remarkably enhanced charge density of inorganic material via regulating contact barrier difference and charge trapping for triboelectric nanogenerators. *Adv Funct Mater*. **2023**;*33*:2304221.
78. Wang C, Guo H, Wang P, Li J, Sun Y, Zhang D. An advanced strategy to enhance TENG output: reducing triboelectric charge decay. *Adv Mater*. **2023**;*35*:2209895.
79. Wen X, Su Y, Yang Y, Zhang H, Wang ZL. Applicability of triboelectric generator over a wide range of temperature. *Nano Energy*. **2014**;*4*:150.
80. Bui VT, Huynh ND, Chau NM, Kim W, Kim H, Oh IK, Huynh DP, Choi D. High-temperature operatable triboelectric nanogenerators using microdome-patterned polyimide for self-powered sensors. *Nano Energy*. **2022**;*101*:107612.
81. McCarty L, Whitesides G. Electrostatic charging due to separation of ions at interfaces: Contact electrification of ionic electrets. *Angew Chem Int Ed*. **2008**;*47*:2188.
82. Nguyen V, Yang R. Effect of humidity and pressure on the triboelectric nanogenerators. *Nano Energy*. **2013**;*2*:604.
83. Guo Y, Wei X, Shu J, Liu B, Yin J, Guan C, Han Y, Gao S, Chen Q. Charge trapping at the MoS_2–SiO_2 interface and its effects on the characteristics of MoS_2 metal-oxide-semiconductor field effect transistors. *Appl Phys Lett*. **2015**;*106*:103109.
84. Conley JF, Lenahan PM. Room temperature reactions involving silicon dangling bond centers and molecular hydrogen in amorphous SiO_2 thin films on silicon. *Appl Phys Lett*. **1993**;*62*:40.
85. Niu SM, Wang SH, Lin L, Liu Y, Zhou YS, Hu YF, Wang ZL. Theoretical study of contact-mode triboelectric nanogenerators as effective power source. *Energy Environ Sci*. **2013**;*6*:3576.
86. Yang B, Zeng W, Peng ZH, Liu SR, Chen K, Tao XM. A fully verified theoretical analysis of contact-mode triboelectric nanogenerators as a wearable power source. *Adv Energy Mater*. **2016**;*6*:1600505.
87. Zhang J, Hu S, Shi Z, Wang Y, Lei Y, Han J, Xiong Y, Sun J, Zheng L, Sun Q, Yang G, Wang ZL. Eco-friendly and recyclable all-cellulose triboelectric nanogenerators and self-powered interactive interface. *Nano Energy*. **2021**;*89*:106354.
88. Zhang H, Quan LW, Chen JK, Xu CK, Zhang CH, Dong SR, Lü CF, Luo JK. A general optimization approach for contact-separation triboelectric nanogenerators. *Nano Energy*. **2019**;*56*:700.
89. Liu X, Yu A, Qin A, Zhai J. Highly integrated triboelectric nanogenerators for efficiently harvesting raindrop energy. *Adv Mater Technol*. **2019**;*4*:1900608.
90. Lin ZH, Cheng G, Lee S, Pradel KC, Wang ZL. Harvesting water drop energy by a sequential contact-electrification and electrostatic-induction process. *Adv Mater*. **2014**;*26*:4690.
91. Choi D, Lee H, Im DJ, Kang IS, Lim G, Kim DS, Kang KH. Spontaneous electrical charging of droplets by conventional pipetting. *Sci Rep*. **2013**;*3*:2037.
92. Ravelo B, Duval F, Kane S, Nsom B. Demonstration of the triboelectricity effect by the flow of liquid water in the insulating pipe. *J Electrostat*. **2011**;*69*:473.
93. Takahashi T. Measurement of electric charge of cloud droplets, drizzle, and raindrops. *Rev Geophys*. **1973**;*11*:903.
94. Cheng G, Lin ZH, Du ZL, Wang ZL. Simultaneously harvesting electrostatic and mechanical energies from flowing water by a hybridized triboelectric nanogenerator. *ACS Nano*. **2014**;*8*:1932.
95. Zheng L, Lin ZH, Cheng G, Wu WZ, Wen XN, Lee SM, Wang ZL. Silicon-based hybrid cell for harvesting solar energy and raindrop electrostatic energy. *Nano Energy*. **2014**;*9*:291.
96. Im B, Lee SK, Kang G, Moon J, Byun D, Cho DH. Electrohydrodynamic jet printed silver-grid electrode for transparent raindrop energy-based triboelectric nanogenerator. *Nano Energy*. **2022**;*95*:107049.
97. Ye CY, Liu D, Peng X, Jiang Y, Cheng RW, Ning CA, Sheng FF, Zhang YH, Dong K, Wang ZL. A hydrophobic self-repairing power textile for effective water droplet energy harvesting. *ACS Nano*. **2021**;*15*:18172.
98. Clanet C, Beguin C, Richard D, Quere D. Maximal deformation of an impacting drop. *J Fluid Mech*. **2004**;*517*:199.
99. Cho H, Chung J, Shin G, Sim JY, Kim DS, Lee S, Hwang W. Toward sustainable output generation of liquid-solid contact triboelectric nanogenerators: the role of hierarchical structures. *Nano Energy*. **2019**;*56*:56.
100. Kim H, Lee C, Kim MH, Kim J. Drop impact characteristics and structure effects of hydrophobic surfaces with micro- and/or nanoscaled structures. *Langmuir*. **2012**;*28*:11250.
101. Liu YH, Moevius L, Xu XP, Qian TZ, Yeomans JM, Wang ZK. Pancake bouncing on superhydrophobic surfaces. *Nat Phys*. **2014**;*10*:515.
102. Tang CL, Qin MX, Weng XY, Zhang XH, Zhang P, Li JL, Huang ZH. Dynamics of droplet impact on solid surface with different roughness. *Int J Multiphase Flow*. **2017**;*96*:56.
103. Bao Y, Wang Z, Shi Z, Gan Y. A modified Weber number capturing the bouncing-wetting transition of droplet impact on rough surfaces. *Adv Mater Interfaces*. **2023**;*10*:2201873.
104. Liu Y, Liu YZ, Chen M. Maximum spreading diameter of bouncing droplets at ultralow Weber numbers. *Langmuir*. **2023**;*39*:7922.
105. Wang X, Fang S, Tan J, Hu T, Chu W, Yin J, Zhou J, Guo W. Dynamics for droplet-based electricity generators. *Nano Energy*. **2021**;*80*:105558.
106. Yun S, Cho S, Kim HW, Cho SB, Lee S, Yong K. Reversible switching performance of water droplet-driven triboelectric nanogenerators using a magnetocontrollable lubricant-infused surface for sustainable power generation. *Nano Energy*. **2022**;*103*:107783.
107. Xing L, Zhang Q, Fu Q, Yu J, Huang X. Preparation and properties of polytetrafluoroethylene superhydrophobic surface. *Colloid Surf A*. **2023**;*670*:131574.
108. Hussain MM, Majjed MK, Ma H, Wang Y, Adil S, Lotfi M. PTFE/EP reinforced MOF/SiO_2 composite as a superior mechanically robust superhydrophobic agent towards corrosion protection, self-cleaning and anti-icing. *Chem Eur J*. **2022**;*28*:202103220.
109. Zhu NX, Wei ZW, Chen CX, Wang D, Cao CC, Qiu QF, Jiang JJ, Wang HP, Su CY. Self-generation of surface roughness by low-surface-energy alkyl chains for highly stable superhydrophobic/superoleophilic MOFs with multiple functionalities. *Angew Chem Int Ed*. **2019**;*58*:17033.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.